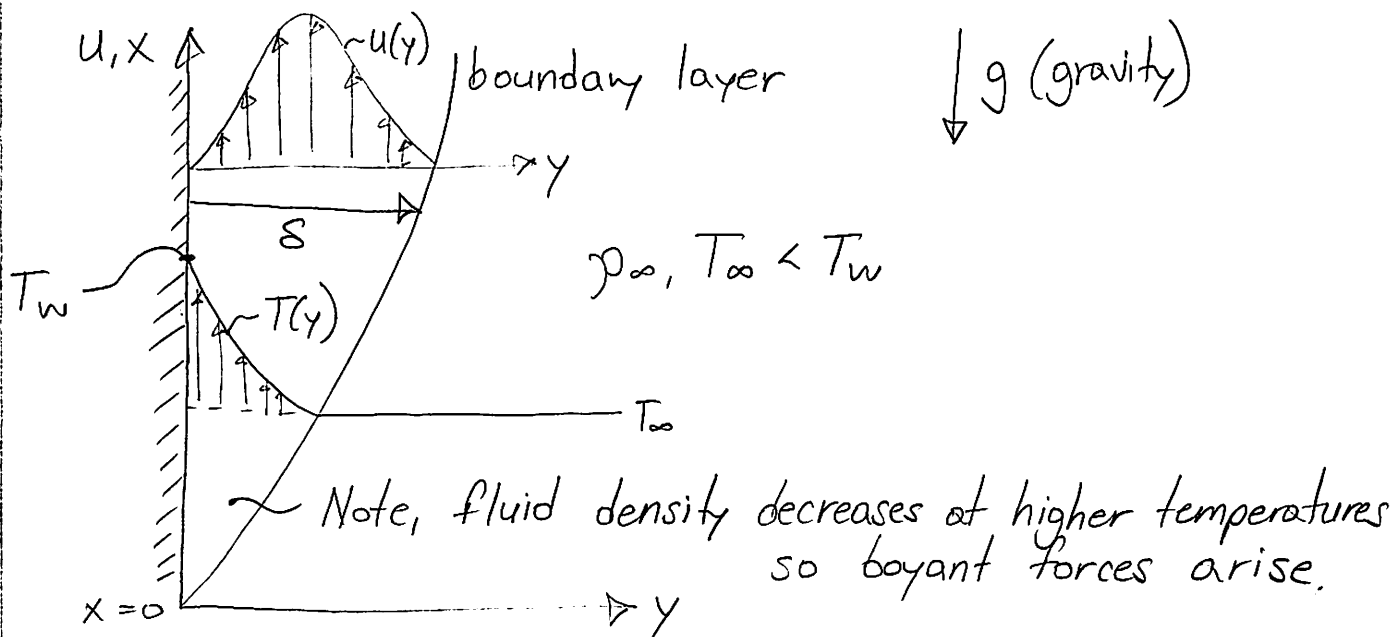


Natural Convection

Unlike forced convection, in which the fluid motion driving force is external to the fluid, natural convection is driven by body forces acting directly on the fluid as a result of heating or cooling.

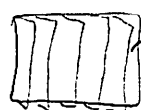
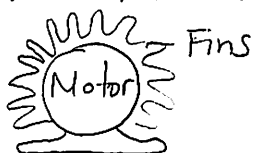
Simplest Case: Hot vertical plane wall



The flow is analogous to flow over a flat plate, but $u = 0$ at $y = 0$ and at $y \rightarrow \infty$.

Natural convection is all around us:

- 1) When humans are stationary, natural convection occurs & we lose heat to the air surrounding us.
- 2) Many heat sinks are designed for natural convection. i.e. transformer cooling, home appliances, induction motors.



To solve, we need to use the momentum & energy equations but now taking gravity into account.

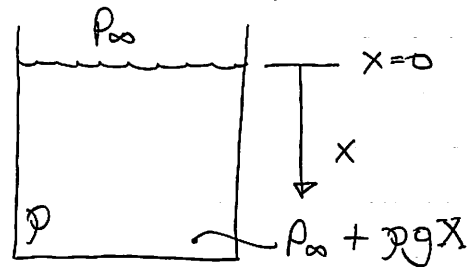
x-momentum:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = - \frac{1}{\rho} \frac{\partial p}{\partial x} + \underbrace{\nu \frac{\partial^2 u}{\partial y^2}}_{\text{Body force (gravity in -x)}} - g \quad (1)$$

To help us solve, we know for a stationary fluid that:

$$p(x) = p_{\infty} + \rho g x$$

(Archimede's principle)



So we can say:

$$\frac{\partial p}{\partial x} = - \rho_{\infty} g$$

density of fluid far away from wall

$$\frac{1}{\rho} \frac{\partial p}{\partial x} = - \frac{\rho_{\infty} g}{\rho} \quad (2)$$

density of fluid close to the wall

Back substitute (2) into (1)

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = g \left(\frac{\rho_{\infty} - \rho}{\rho} \right) + \nu \frac{\partial^2 u}{\partial y^2} \quad (3)$$

However we need the relation between density ρ and temperature, T . We can use the following:

$$\beta = \frac{1}{V} \frac{\partial V}{\partial T} \bigg|_p$$

$V \equiv$ fluid specific volume
 $\beta \equiv$ coefficient of volumetric thermal expansion

$$\beta = - \frac{1}{\rho} \frac{\partial \rho}{\partial T} \bigg|_p \quad (4)$$

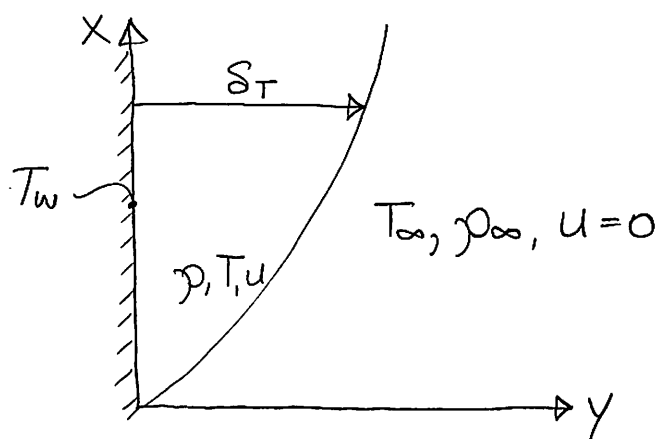
$$\Rightarrow \frac{\rho_{\infty} - \rho}{\rho} = \beta (T - T_{\infty})$$

So now our momentum equation becomes:

$$\underbrace{u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y}}_{\text{Inertia Forces}} = \underbrace{g\beta(T-T_\infty)}_{\text{Buoyancy Forces}} + \underbrace{\nu \frac{\partial^2 u}{\partial y^2}}_{\text{Viscous Forces}}$$

Let's try to solve this using scaling analysis:

Looking at the problem as a transient conduction prob.



$$\delta_T \sim \sqrt{\alpha t} \quad ; \quad \begin{array}{l} t \equiv \text{time} \\ \alpha \equiv \text{fluid thermal diffusivity of the fluid} \end{array}$$

Speed of heat conduction (page 65 of the notes)

So what is t here? Well, we know $t = \frac{x}{u}$ $\leftarrow$ location on plate
 $\leftarrow$ flow speed

$$\delta_T \sim \sqrt{\frac{\alpha x}{u}} \quad \text{so} \quad u \sim \frac{\alpha x}{\delta_T^2} \quad \text{and} \quad y \sim \delta_T$$

So let's break down the scaling of the forces

Inertia, I	Viscosity, V	Boyancy, B
$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \sim \frac{u^2}{x}$ $\sim \frac{\alpha^2 x}{\delta_T^4}$	$\nu \frac{\partial^2 u}{\partial y^2} \sim \nu \frac{u}{y^2}$ $\sim \frac{\nu \alpha x}{\delta_T^4}$	$g\beta(T-T_\infty)$ $= g\beta\Delta T$

Now if we take ratios of our forces to see which ones dominate: (focus on boyancy)

I/B	V/B	B/B
$\frac{\alpha^2 x}{\delta_T^4 g\beta\Delta T}$	$\frac{\nu \alpha x}{\delta_T^4 g\beta\Delta T}$	1

Assuming that $V/B \sim O(1)$ (Inside the natural convection boundary layer). Think of the boundary layer definition.

$$\frac{\nu \alpha x}{\delta_T^4 g\beta\Delta T} \sim 1$$

Multiplying both sides by $\frac{x^3}{x^3}$

$$\frac{\nu \alpha}{g\beta\Delta T x^3} \left(\frac{x}{\delta_T}\right)^4 \sim 1$$

$$\boxed{\frac{x}{\delta_T} \sim \left(\frac{g\beta\Delta T x^3}{\nu \alpha}\right)^{1/4}} \Rightarrow (\text{Rayleigh number})^{1/4}$$

$$Ra_x = \frac{g\beta\Delta T x^3}{\alpha U} \quad ; \quad \frac{x}{\delta_T} \sim Ra^{1/4}$$

$$Gr_x = \frac{g\beta\Delta T x^3}{U^2} \Rightarrow \text{Grashof number}$$

$$Ra = \underbrace{\frac{\text{Buoyancy Force}}{\text{Viscous Force}}}_{Gr} \cdot \underbrace{\frac{\text{Momentum Diffusivity}}{\text{Thermal Diffusivity}}}_{Pr \text{ (Prandtl \# = } \frac{U}{\alpha})} = Gr \cdot Pr$$

If: $Ra < Ra_{crit}$, conduction dominates ($Gr \ll 1, Pr \ll 1$)
 $Ra > Ra_{crit}$, convection dominates ($Gr > 1, Pr > 1$)

Think of Rayleigh # & Grashof # as an analogy to Reynolds #, but here instead of inertia, we have buoyancy.

Heat Transfer (Natural Convection)

$$q''|_{y=0} = -k_f \left. \frac{\partial T}{\partial y} \right|_{y=0} \sim h \frac{\Delta T}{\delta_T}$$

$$\left(\frac{q''|_{y=0}}{\Delta T} \right) \cdot \frac{x}{k_f} = \frac{h_x x}{k_f} = Nu_x \sim \frac{x}{\delta_T} \sim Ra^{1/4}$$

$$\boxed{Nu_x \sim Ra_x^{1/4}} \Rightarrow \text{Scaling tells us this is what we expect.}$$

Full analytical result: $\boxed{Nu_x = 0.394 Ra_x^{1/4}} \Rightarrow T_w = \text{constant}$

$$\boxed{\overline{Nu_L} = \frac{\bar{h} L}{k_f} = 0.525 Ra_L^{1/4}} \Rightarrow \boxed{\overline{Nu_{L,exp}} = 0.52 Ra_L^{1/4}}$$

For cases where $T_w \neq \text{constant}$ & $q'' = \text{constant}$

We can define a modified Rayleigh number, Ra_x^*

$$Ra_x^* = \frac{g\beta(q''/k_f)x^4}{\nu\alpha} \Rightarrow \text{Constant heat flux Rayleigh \#}$$

$$\left(\frac{q''|_{y=0}}{\Delta T}\right) \cdot \frac{x}{k_f} = \frac{h_x x}{k_f} = Nu_x$$

$$Nu_x = 0.503 Ra_x^{*1/5} \Rightarrow \text{Constant heat flux natural convection, vertical wall Analytical result.}$$

In general: $Ra_L > 10^9 \Rightarrow \text{Turbulent free convection}$

$$\overline{Nu_L} = \frac{\overline{h}L}{k_f} = 0.1 Ra_L^{1/3} ; 10^9 < Ra_L < 10^{13}$$

For laminar flow:

$$\overline{Nu_L} = \frac{\overline{h}L}{k_f} = 0.59 Ra_L^{1/4} ; 10^4 \leq Ra_L < 10^9$$

(Similar to what we derived)

Note, both correlations above are for constant vertical wall temperature ($T_w = \text{constant}$).

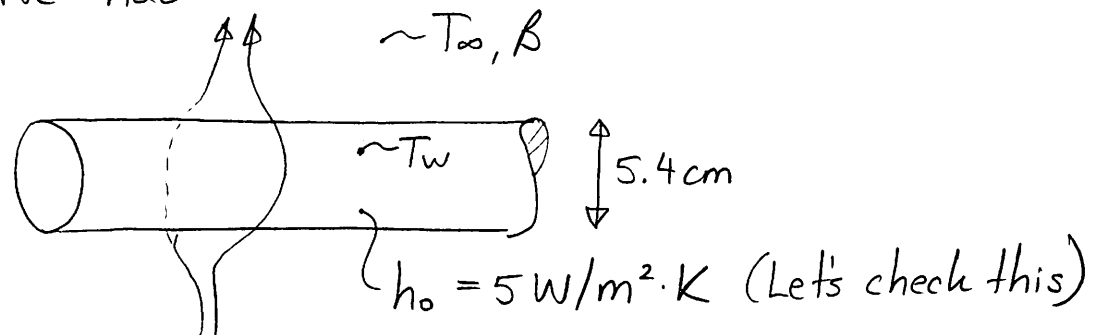
For other cases; i) Inclined walls
ii) Channels
iii) Enclosures
iv) Cylinders & Spheres

} See Chapter 9 of the textbook Incropera.

* All properties evaluated at $\overline{T} = \frac{T_w + T_\infty}{2}$; $\beta = \frac{1}{T}$ for ideal gas. (154)

Example | Let's check the validity of our previous example with New England pipes.

We had:



First calculate the Rayleigh #:

$$Ra_D = \frac{g\beta\Delta T D^3}{\alpha\nu} \quad \text{note, we use } D \text{ here instead of } L.$$

For air at $T = \frac{-10^\circ\text{C} + 1^\circ\text{C}}{2} = 4.5^\circ\text{C}$ (Table: A.4)

$$\rho_{\text{air}} = 1.2 \text{ kg/m}^3$$

$$\alpha_{\text{air}} = 20 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\nu_{\text{air}} = 14 \times 10^{-6} \text{ m}^2/\text{s}$$

$$k_{\text{air}} = 24 \times 10^{-3} \text{ W/m} \cdot \text{K}$$

$$\beta = \frac{1}{T_{\infty}} = \frac{1}{263.15 \text{ K}} = 0.0038 \text{ K}^{-1}$$

$$Ra_D = \frac{(9.81 \text{ N/kg})(0.0038 \text{ K}^{-1})(11^\circ\text{C})(0.054 \text{ m})^3}{(20 \times 10^{-6} \text{ m}^2/\text{s})(14 \times 10^{-6} \text{ m}^2/\text{s})} = 2.3 \times 10^5$$

Looking at Eq. 9.33 of Textbook & corresponding Table 9.1

$$\overline{Nu}_D = \frac{\overline{h}D}{k_f} = 0.48 Ra_D^{1/4} = 0.48 (2.3 \times 10^5)^{1/4} = 10.51$$

$$\overline{h} = \frac{\overline{Nu}_D \cdot k_f}{D} = \frac{(10.51)(0.024 \text{ W/m} \cdot \text{K})}{0.054 \text{ m}} = 4.67 \text{ W/m}^2 \cdot \text{K} \quad (155)$$